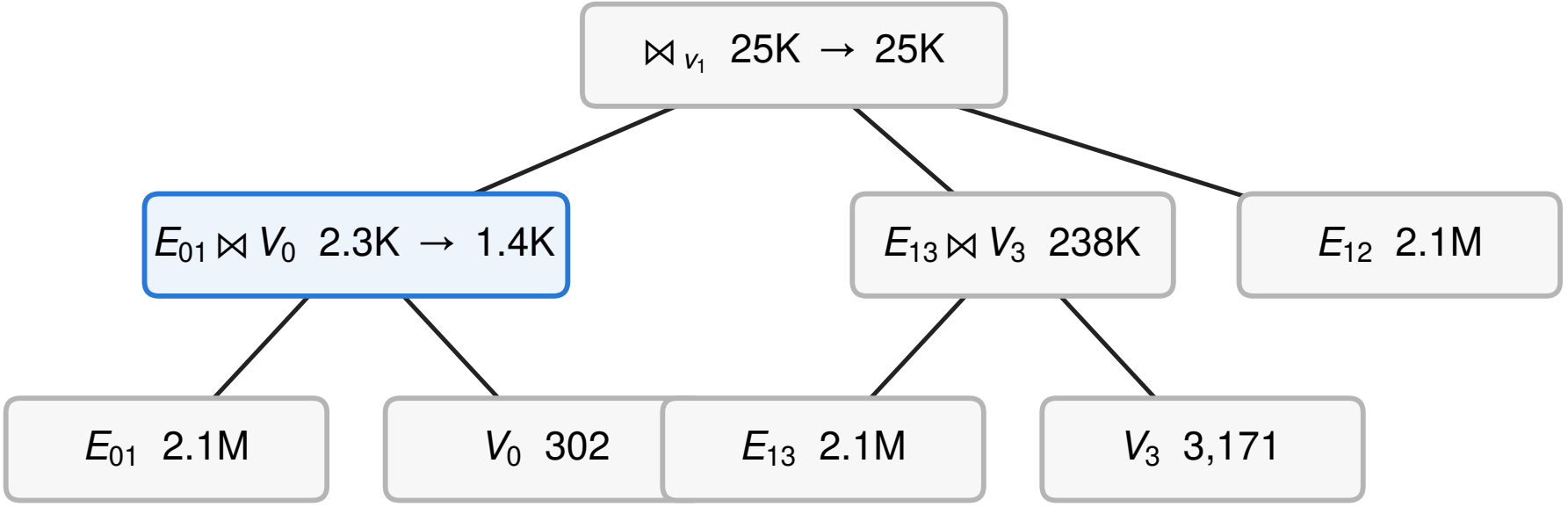
a host join tree of the example: rows per node, before → after $v_3 \to v_1$



benefit = rows removed below the root from nodes with v_1: 879 of 2.3K;
cost = a pull over B_1, up to 63K entries: not admitted. The root is output,
and $E_{13} \bowtie V_3$ joins v_1 through the transfer's own edge and keeps its rows